TreeDiff: AST-Guided Code Generation with Diffusion-based LLMs

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First AST-aware masking for DLLMs → **+13.3% on HumanEval+**, narrows gap with autoregressive models

Motivation

Diffusion-based LLMs (DLLMs) struggle with **syntactic precision** in code generation. Standard random token masking:

- Ignores syntactic boundaries during denoising
- Fails to capture long-range hierarchical dependencies
- Treats all tokens equally — but they are not

CORE INSIGHT

"For code, noise should not be purely stochastic — structural priors enable DLLMs to learn program-level dependencies."

Method: AST-Guided Masking

TreeDiff parses code into an **Abstract Syntax Tree (AST)** and uses it to guide which tokens to mask — targeting structurally critical nodes.

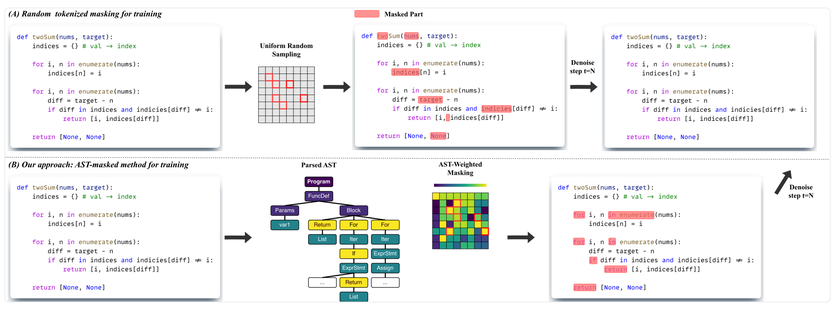


Figure 1: Uniform Random Masking (top) vs. AST-Weighted Sampling Masking (bottom).

Region-specific Corruption

Prompt	Kept intact (semantic conditioning)
Reasoning	Random token masking
Code	AST-weighted masking

GOAL

Preserve semantic conditioning while injecting structural noise into code regions.

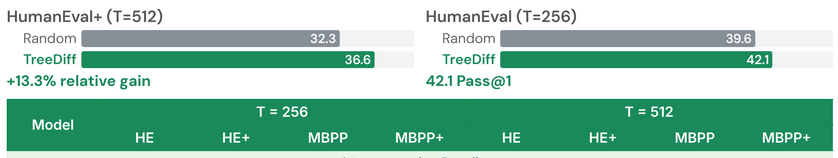
AST Node Weighting

Category	Examples	Weight	Purpose
Skeletal	Imports, Constants	0.15	Preserve skeleton
Data Flow	Assigns, Calls	0.42	Moderate priority
Conditionals	if, elif, else	0.58	Core logic
Control Flow	for, while, return	**0.60**	Highest priority

Main Results (Pass@1 %)

13.3%
Relative Gain
HumanEval+ (T=512)

42.1%
HumanEval Pass@1
(vs 39.6% baseline)



Model	T = 256				T = 512			
	HE	HE+	MBPP	MBPP+	HE	HE+	MBPP	MBPP+
Autoregressive Baselines								
DeepSeek-Coder-33B	50.6	44.5	80.4	70.1	50.6	44.5	80.4	70.1
CodeQwen... 7B	51.8	45.7	73.5	60.8	51.8	45.7	73.5	60.8
CodeLlama-13B	42.7	38.4	63.5	52.6	42.7	38.4	63.5	52.6
Diffusion-based LLMs (LLaDA Backbone)								
LLaDA-Instruct	39.6	34.8	51.9	43.1	36.6	31.7	39.4	31.7
+ Random Masking	39.6	35.4	52.9	43.9	38.4	32.3	41.5	32.8
Ours								
+ TreeDiff	**42.1**	**37.2**	**52.9**	**44.2**	**40.2**	**36.6**	**42.3**	**33.3**

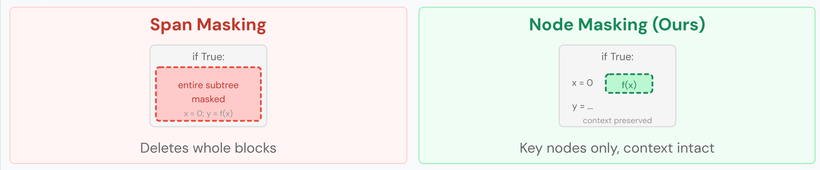
Source: AR baselines from EvalPlus Leaderboard. HE = HumanEval; HE+ = HumanEval+.

Ablation: Masking Granularity

HumanEval (HE / HE+)		
Strategy	**T=256**	**T=512**
Random Masking	39.6 / 35.4	38.4 / 32.3
AST (Span)	40.9 / 36.6	40.2 / 36.6
AST (Node)	**42.1 / 37.2**	**40.2 / 36.6**

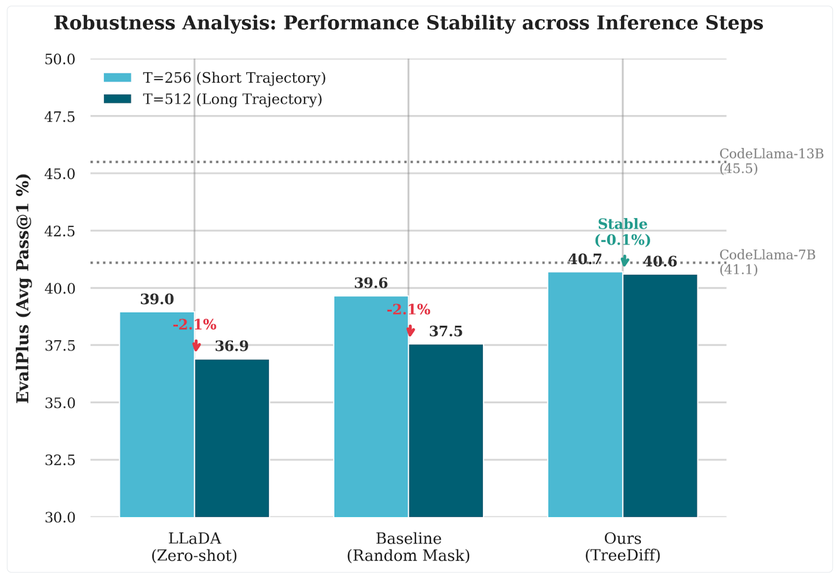
MBPP (MBPP / MBPP+)		
Strategy	**T=256**	**T=512**
Random Masking	52.9 / 43.9	41.5 / 32.8
AST (Span)	51.6 / 42.9	39.7 / 31.5
AST (Node)	**52.9 / 44.9**	**42.3 / 33.3**

Why Node-level > Span-level?



Robustness Analysis

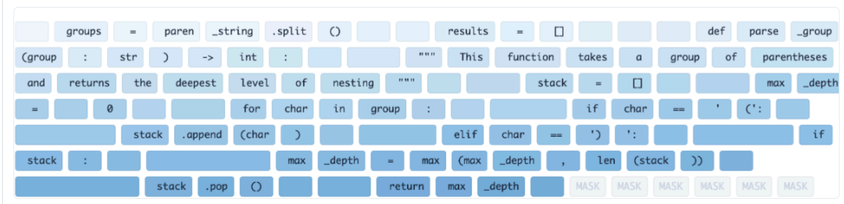
Random masking accumulates syntactic/logical drift at T=512. TreeDiff remains stable (-0.1%) by anchoring control-flow and skeleton nodes.



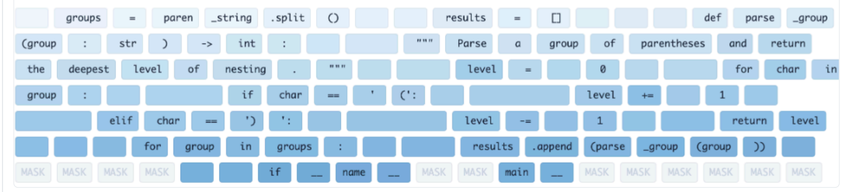
TreeDiff maintains stable performance as T increases 256 to 512, while baselines degrade by ~2.1%.

Generation Trajectory Analysis

Task: Max nesting depth of parentheses (HumanEval/6), Step 110/256.



Random Mask Baseline: Uses shallow counter heuristic, fails on nested structures. Fragmented generation.



TreeDiff: Establishes stack-based control flow (stack.pop()) early. Correctly captures recursive nesting.

Generated Code Comparison

HumanEval/6 — Nested parentheses depth:

TreeDiff

```
for char in group:
    if char == '(':
        stack.append(char)
        max_depth = max(
            max_depth, len(stack)
        )
    elif char == ')':
        if stack: stack.pop()
```

Random Masking

```
count = 0
for char in group:
    if char == '(':
        count += 1
    # Fails to track
    # nested peak depth
```

Why TreeDiff Works

- Random masking **breaks syntactic anchors** — the model loses structural landmarks
- Span masking **removes too much local context** — entire subtrees vanish
- TreeDiff **masks high-influence AST nodes** while preserving surrounding code structure

Key Contributions

- **First AST-aware masking** for DLLMs in code generation
- **13.3% relative improvement** on HumanEval+ vs random masking
- Trained on **150K** long code reasoning samples
- Negligible overhead: AST parsing is O(L)

Conclusion

TreeDiff addresses a fundamental limitation: **random masking ignores code structure**. By aligning corruption with AST hierarchy, the model learns programming language compositionality, reconstructing programs that respect grammatical boundaries and capture long-range dependencies.

Impact: Structural priors unlock DLLMs for code — TreeDiff narrows the gap between diffusion and autoregressive paradigms.

Setup: LLaDA-8B backbone · 8x NVIDIA A100 · LoRA (r=64, alpha=128) · 4096 max tokens · LR 5e-5 · Benchmarks: HumanEval/+, MBPP/+

Limitations & Next Steps

- Long iterative denoising can be slow for very long traces (capped at 1024 tokens)
- Current setup uses LoRA due to 8xA100 memory constraints
- Next: faster inference schedules, broader language coverage, stronger backbones

Experimental Setup

Backbone	LLaDA-8B-Instruct
Training Data	150K samples (OpenCodeReasoning)
Max Seq Length	4,096 tokens
GPUs	8× NVIDIA A100
Adaptation	LoRA (r=64, α=128)
Benchmarks	HumanEval/+, MBPP/+

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